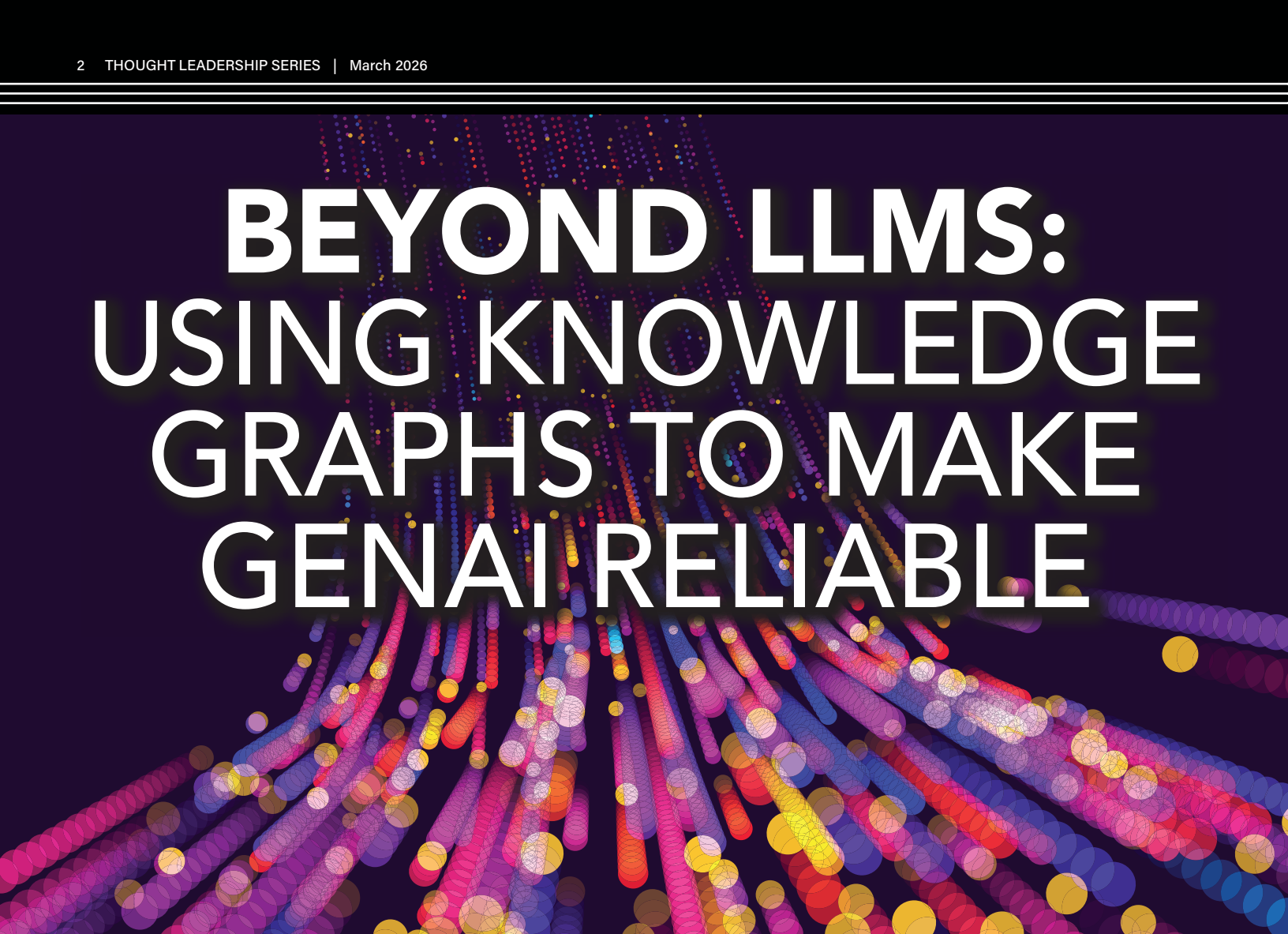


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BEYOND LLMS: USING KNOWLEDGE GRAPHS TO MAKE GENAI RELIABLE

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BEYOND LLMs: USING KNOWLEDGE GRAPHS TO MAKE GENAI RELIABLE

GenAI adoption is accelerating across the enterprise, but business value is falling behind. Analysts predict that 60% of AI initiatives will be abandoned without AI-ready data, and as many as 70-85% will never reach production. Concerns over hallucinations and unreliable output continue to grow.

The limitation is not model intelligence but missing context. Studies show that grounding LLMs in knowledge graph structures can improve answer accuracy, signaling a turning point: Successful GenAI depends on connected data and relationships, not language generation alone.

With enterprise data existing in silos across multiple systems and formats, it is difficult to integrate into a cohesive data architecture for the underlying LLM to work with. Due to data silos, the model also cannot provide causal reasoning for its output, resulting in low data utilization that renders GenAI systems unexplainable, and at best, “black-boxed.”

The synergy between knowledge graphs (KGs) and AI offers benefits in grounding

and validating LLM responses, leading to more accurate, contextually relevant, and trustworthy AI-generated content.

LLMs, when left to their pre-trained parameters alone, often generate “believable-sounding but factually incorrect” responses. By grounding LLM outputs in the information stored in KGs, the risk of hallucination is significantly reduced. This is especially valuable in domains where accuracy is paramount, such as healthcare or technical support.

WHAT ARE KNOWLEDGE GRAPHS?

A knowledge graph (KG) stores a knowledgebase using graph-based data structures to represent and store the underlying information. The nodes of the graph represent entities, such as objects, places, people, etc. The edges represent the relationships (such as “is a customer,” “was born in,” etc.) between different entities.

KGs are also called semantic networks because they store information about how different entities are related to each other.

KNOWLEDGE GRAPHS AS THE FOUNDATION FOR GENAI

While knowledge graphs do the patching, the elemental data must still be relevant and high-quality. Enterprises need to lay the groundwork by identifying the right datasets for use cases and consolidating them across modalities and formats into a cohesive data architecture.

As with any other AI model training, enterprises must invest in building data pipelines, defining the ontology or taxonomy, and determining the foundational relationships or domains under which they rest.

Doing this on an enterprise scale can be complex, and it becomes even more so when LLMs need to “self-learn” with dynamic knowledge, which requires constantly adding new data sources to the underlying data architecture.

As per Dattaraj Rao at Persistent Systems, knowledge graphs have the potential to trigger GenAI-powered business transformation with:

- **Enhanced data utilization:** Since knowledge graphs integrate across siloed enterprise datasets, they enable the overlaid LLM to identify synergies or gaps and extract semantic relationships between various entities to provide a richer and more contextual output while explaining the reasoning or underlying causality. A loosely knotted data architecture improves data utilization, helping enterprises derive deeper insights from disparate data sets.
- **Rapid inference:** Since a knowledge graph represents data as nodes and associated dependencies as edges, it is easier for the LLM to make faster inferences that cut across datasets—borne out of a deeper understanding of the variables that interact with and influence each other. It becomes easier for the model to underline its output with more context, identify relationships that otherwise were hidden, and offer analysis efficiently.
- **Accelerated model deployment:** As a platform offering richer data relationships across enterprise data sets, knowledge graphs can train multiple LLMs across use cases without the heavy lifting required for individual models.
- **Improved GenAI adoption:** Knowledge graphs can ground LLM output in responsible AI principles so that they handle more complex queries about information stored in disparate datasets. They can also back up the answers with context, facts, and underlying relationships that help non-technical users make sense of highly technical information—thereby boosting adoption.
- **Better collaboration:** With a bird's-eye view, LLMs can enable enterprise users with ready insights that foster enhanced cross-departmental collaboration, speed up innovation, enable accurate prediction models, empower teams to make more informed decisions, and instill better alignment between adjacent teams.

BEST PRACTICES FOR MAINTAINING THE COMBINATION OF KNOWLEDGE GRAPHS AND LLMs

According to DataCamp, it is necessary to continually update the graph database with the latest information. An outdated KG results in the LLM giving outdated answers.

Query optimization involves rewriting database queries and using the right indexes to make database access more efficient. Steps

include indexing, query parametrization, asynchronous querying, and load balancing.

LLMs are memory-heavy models and often need expensive hardware to run on. It is helpful to reduce the size of AI models using techniques including LLM Distillation and Quantization.

LLM Distillation is a technique that uses a smaller model, called the “student model,” to replicate the responses of a larger (teacher) model. Only the teacher model is trained on the original dataset, and the student model is trained only to match the teacher’s responses on many inputs. This process is called knowledge distillation, and it is cheaper than training a model from scratch. However, the student’s performance may not be reliable for new queries on which it has not been trained.

Quantizing LLMs reduces the precision of the model weights. The weights of base AI models are typically represented as 32-bit or 16-bit floating-point numbers. It has been observed that reducing the precision of model weights to 8-bit integers (in some cases, even lower to 1-bit) results in comparable performance at a fraction of the size and cost.

It is essential to periodically re-evaluate the LLM’s performance after deployment. It might prove necessary to fine-tune the model on an updated dataset, use an updated version of the model, or even deploy a new model. Tools such as MLFlow can help evaluate LLM performance in a variety of tasks.

And fine-tuning the LLM on a domain-specific dataset enhances its reasoning abilities in that domain and enables it to construct sentences using the right structure and jargon.

EXAMPLES OF EVERYDAY USE CASES

KGs can provide a structured and up-to-date repository of knowledge that a chatbot can then programmatically access to get current information and provide more relevant responses to user queries.

The KG supplements the chatbot’s default knowledge, which is based on the training dataset. The chatbot then learns how to construct sentences based on the training dataset and acquires application-specific information from the KG.

LLMs have been also used as recommendation engines. For example, an LLM can analyze an article’s content and recommend other relevant articles the reader may be interested in. LLMs are also used to create interactive systems where the user can “talk to” the recommendation engine.

KGs can organize and store information on user behavior, interests, interactions, and more. LLMs can analyze this information and use it to generate tailored responses. Users can then receive personalized recommendations based on their individual preferences.

In knowledge-based domains such as medicine, finance, and law, LLMs combined with KGs improve the ease of accessing information.

In medicine, they can suggest possible diagnoses based on patient symptoms and history. In finance, they supplement analyst efforts by making accessing relevant information from financial reports easier. In law, they can aid lawyers by assisting with retrieving relevant past cases and judgments and providing summaries of large legal documents.

Combining large language models with knowledge graphs is critical in real-time applications including news aggregation and stock market analysis.

KGs are efficient at processing and organizing new information in real time. They can provide the informatics backend for LLMs to handle applications based on current affairs (news aggregation) and rapidly changing environments (e.g., the stock market).

CONCLUSION

When organizations deploy LLMs without grounding them, projects can fail because power without foundation is unreliable. When done properly by combining technologies that complement each other, these systems will genuinely augment human intelligence.

As enterprises move from experimentation to production AI, success will depend less on the sophistication of the model and more on the quality and structure of the data that supports it.

Knowledge graphs provide the contextual foundation that LLMs lack on their own, connecting entities, relationships, and meaning across complex data environments.

By combining the reasoning capabilities of LLMs with the structured intelligence of knowledge graphs, organizations can build GenAI systems that are more accurate, explainable, and trustworthy. In the emerging AI landscape, the future of reliable GenAI will be built not just on language models, but on connected knowledge. ■

—Stephanie Simone

Beyond the Model: Why Reliable GenAI Depends on Context, Control, and Connected Data



Generative AI (GenAI) has captured enterprise attention faster than almost any technology shift in recent memory. Yet for many organizations, the gap between excitement and value is becoming harder to ignore. Pilots are easy to launch. Demos are easy to impress with. Production-grade outcomes are much harder to achieve.

That is because the biggest challenge in enterprise AI is rarely the model itself. It's the environment around it.

Large language models (LLMs) are powerful pattern engines. They can summarize, classify, draft, and converse with astonishing fluency. But they do not inherently understand your business, your policies, your terminology, or the relationships that give enterprise information its meaning. Without that context, even advanced models can produce outputs that sound persuasive while still being incomplete, inconsistent, or wrong.

This is where many AI initiatives begin to stall. Organizations invest in model access, prompting strategies and interfaces, only to discover that reliability does not emerge from language alone. It comes from grounding. It comes from governance. And increasingly, it comes from connected data structures that help AI operate with more context and precision.

That is why knowledge graphs are moving to the center of serious enterprise AI conversations.

A knowledge graph does more than store information; it connects entities, concepts, documents, and relationships in a way that reflects how organizations actually work. Customers connect to products, products connect to contracts, contracts connect to obligations, obligations connect to policies, and policies connect to decisions. Instead of retrieving isolated fragments of content, AI can work from a richer map of meaning.

This matters because enterprise trust is built on more than retrieval. It's built on whether an answer can be traced, explained, and defended.

In regulated and knowledge-intensive industries, that standard is not optional. A legal assistant must cite the right sources and respect precedent. A pharmaceutical research workflow must connect scientific findings, trial data, and compliance requirements. A financial services application must align recommendations to policy, suitability, and auditability. In each case, the issue is not

whether AI can generate an answer—it's whether the organization can stand behind it.

Knowledge graphs help make that possible by organizing information around meaning, not just keywords. They improve disambiguation, strengthen retrieval, and expose relationships that would otherwise remain hidden across silos. They also make it easier to understand why a system produces a particular answer, which is essential when trust, compliance, and accountability are on the line. This becomes even more powerful when combined with rules-based governance.

For all the attention given to model scale, enterprise value increasingly depends on something more practical: architectures that combine neural and symbolic approaches. Neural AI brings flexibility, fluency, and pattern recognition. Symbolic approaches bring rules, policy enforcement, explainability, and control. Knowledge graphs provide the connective layer that links data, content, and business context together.

This combination creates a stronger foundation for enterprise AI. It allows organizations to move beyond hoping a prompt will produce the right output and toward systems designed to be dependable. It enables teams to ask not only, "Can the model answer this?," but also, "Can we verify it, govern it, and improve it over time?"

*At Progress®, we often describe this with a phrase that can sound provocative at first: **Make AI Boring.***

What we mean by *boring* is simple. In business, the most valuable AI is not the most theatrical; it's the most reliable. It produces answers grounded in trusted enterprise data. It works within clear guardrails. It reflects the language and logic of the business. It can be monitored, governed, and scaled.

This is the real shift now underway—enterprise AI is moving beyond fascination with model capability alone and toward the harder, more important work of building systems organizations can trust in practice.

Knowledge graphs are not a side component in that future—they are becoming foundational to it. Because when AI moves from experimentation to decision support, context is no longer optional. It's the difference between an answer that is merely impressive and one that is truly trustworthy. ■

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